

Alpine Glacier Dynamics

A valley glacier flows downhill under its own weight. The driving stress is a function of the surface slope, the ice thickness and the gravitational acceleration, and it is resisted by basal friction and internal deformation.

Ice deforms by dislocation creep, in which crystals deform plastically over centuries, and by recrystallisation, which is faster and dominates at depth. The resulting velocity profile is close to linear with depth near the bed and steepens upward.

A glacier advances when accumulation in the accumulation zone exceeds ablation below the equilibrium line. A surge is a short episode of dramatic acceleration caused by water reaching the bed, which decouples the ice from its substrate.

Moraine Deposition and Landforms

Material transported at the glacier surface is supraglacial debris, while debris carried at or within the base is englacial. When a glacier retreats, it leaves lateral moraines marking former ice margins and a terminal moraine at the snout.

Till is the unsorted, unstratified sediment deposited directly from the ice. Its matrix-supported, poorly sorted texture is diagnostic of direct deposition, in contrast to outwash, which is rounded, sorted and bedded because meltwater reworked it.

Drumlins are streamlined hills of till whose long axis records the direction of ice flow. They are common in areas of former continental ice sheets and are useful for reconstructing palaeo-ice flow even where the ice itself is long gone.